D 11758 (Pages : 4) Name.....

Reg. No.....

THIRD SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION NOVEMBER 2021

B.Com.

BCM 3A 11—BASIC NUMERICAL SKILLS

(2014—2016 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each carries 1 *mark.*

I. Choose the correct answer :

1 $ax^2 + c = 0$ is a ————————.

(a) Simple linear equation. (b) Simultaneous equations.

(c) Quadratic equations. (d) Differential equation.

2 $A \cap B' =$ ————————.

(a) A. (b) B'.

(c) A – B. (d) A – B'.

3 Which of the following statement is wrong ?

(a) Mean is rigidly defined.

(b) Mean has some mathematical properties.

(c) Mean is not affected due to sampling fluctuations.

(d) Mean is not affected by extreme values.

4 Statistics is defined in terms of numerical data in the :

(a) Singular sense. (b) Plural Sense.

(c) Both. (d) None.

5 Which one is an absolute measure of dispersion ?

(a) Range. (b) Quartile deviation.

(c) Standard deviation. (d) None.

Turn over

II. Fill in the blanks :

6 How many terms are there in 20, 25, 30, ———— 140 ?

7 Given the term in the sequence 1, 3, 7, 15, 31, next term is ————.

8 ———— is defined as the set of all possible subsets of a given set.

9 ———— refers column headings.

10 ———— is one of the solutions to the equation $3x^2 - 4x + 1 = 0$.

(10 × 1 = 10 marks)

Part B

Answer any **eight** *questions.*
Each question carries 2 *marks.*

11 If A = $\{1,2, 3, 4, 5\}$ B = $\{4, 5, 6, 7\}$ C = $\{1, 3, 6, 7\}$. Prove that :

$(A - B) \cap (A - C) = A - (B \cup C)$.

12 Represent $(A \cap B)'$ by using Venn diagram.

13 If $A = \{1, 2\}$ $B = \{4, 5\}$. Find A × B.

14 Find A – 3B :

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & -1 & 5 \end{pmatrix} B = \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}.$$

15 Find the value of determinants :

$$A = \begin{vmatrix} 1 & 2 & -3 \\ 2 & -1 & 2 \\ 3 & 2 & 4 \end{vmatrix}.$$

16 Solve the equation $x^2 - 4x + 3 = 0$.

17 Solve $4x + 2y = 6$
$5x + y = 6$.

18 What is geometric progression ?

19 Find the rate of interest per annum if the simple interest on a principal of Rs. 5000 is 800 for 4 years.

20 The mean mark obtained by 50 boys in a class is 40 and mean marks of 30 girls of the same class is 46. Find the mean of the marks of all the students taken together.

(8 × 2 = 16 marks)

Part C

Answer any **six** *questions.*
Each carries 4 *marks.*

21 Find median from the following :

Wage	10	12	15	18	20	25	30
Frequency	3	5	8	12	13	12	7

22 For the following data calculate standard deviation and its co-efficient of variation :

Marks	:	2	4	6	8	10
Number of students	:	8	10	16	9	7

23 Compute Quartile Deviation from the following data :

Class	0-40	41-80	81 - 120	121 - 160	161-200	201 - 240	241 & Above
Frequency	394	461	391	334	169	113	148

24 Explain the steps in the construction of index numbers.

25 Explain different types of diagrams used for the presentation of data.

26 Solve $X^2 - 4x = -3$.

27 By using Venn diagram prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

28 Find the sum of 8 terms of the GP 1, 3, 9, 27,.....

(6 × 4 = 24 marks)

Part D

Answer any **two** *questions.*
Each carries 15 *marks.*

29 Out of the total 200 students who appeared for B.Com examination from a centre, 95 failed in accounts, 100 failed in statistics, and 120 in costing. Those who failed both in accounts and statistics were 80, those who failed both in statistics and costing were 82 and those who failed both accounts and costing were 85. The students who failed in all subjects were 75. Find out the number who failed at least in any one of the subjects.

Turn over

30 Solve the following equations by using matrices :

$5x - 6y + 4z = 15$
$7x + 4y - 3z = 19$
$2x + y + 6z = 46.$

31 Calculate standard deviation and co-efficient of variation from the following data :

Class	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	6	8	10	15	12	19

(2 × 15 = 30 marks)